

Multiple Choice

1. **Answer:** A

Options: A) a circle; B) an infinite interval; C) the negative real axis; D) the positive real axis; E) a ray

Note: Correct option: A - a circle

2. **Answer:** A

Options: A) \$125.00; B) \$20.70; C) \$130.00; D) \$133.00; E) \$115.00

Note: Correct option: A - \$125.00

3. **Answer:** B

Options: A) 11.234567; B) 10.065778; C) 6.789012; D) 7.654321; E) 9.876543

Note: Correct option: B - 10.065778

4. **Answer:** A

Options: A) 16 over 36; B) 9 over 4; C) 8 over 18; D) 48 over 108; E) 24 over 54

Note: Correct option: A - 16 over 36

5. **Answer:** A

Options: A) 9; B) 7; C) 8; D) 1; E) 4

Note: Correct option: A - 9

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '24'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.4295'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '56'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** If two numbers give the same remainder when divided by 5, they are said to be equivalent, modulo 5. From n^2 to n^3 , we have multiplied by n . Since n^2 is equivalent to 4 (modulo 5), and n^3 is equivalent to 2 (modulo 5), we are looking for an integer n for which $4 \cdot n$ is equivalent to 2, modulo 5. Notice that if n is greater than 4, then we can replace it with its remainder when divided by 5 without changing whether it satisfies the condition. Therefore, we may assume that $0 \leq n < 5$. Trying 0, 1, 2, 3, and 4, we find that only 3 times 4 leaves a remainder of 2 when divided by 5.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 16. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.